

VOROGOVO, Russian Federation

WMO: 239730

Lat: 61.031N

Lon: 89.614E

Elev: 159

StdP: 14.61

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-46.4	-40.6	-49.1	0.3	-44.5	-44.6	0.4	-39.8	24.7	11.3	22.2	2.7	2.6	0	0.682

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.3	82.9	65.7	79.6	64.5	76.1	63.3	68.8	78.0	67.0	75.4	65.0	72.8	6.2	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.6	95.2	72.8	63.7	88.8	71.3	61.8	83.0	69.3	33.2	78.4	31.7	75.5	30.2	72.7	79.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.4	17.7	15.5	-48.5	88.3	6.9	2.9	-53.4	90.4	-57.4	92.1	-61.3	93.7	-66.2	95.8		
(5)			-46.7	71.9	5.1	2.8	-50.4	73.9	-53.4	75.5	-56.2	77.1	-59.9	79.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	27.9	-10.1	-2.6	14.4	28.9	42.9	59.7	65.0	58.1	45.6	29.6	6.9	-5.9	(6)
(7)		DBStd	28.63	17.95	15.32	12.79	9.93	9.40	8.82	6.37	6.93	7.63	10.44	16.97	18.00	(7)
(8)		HDD50	9206	1864	1472	1103	634	265	21	0	11	175	634	1294	1733	(8)
(9)		HDD65	13697	2329	1892	1568	1084	687	201	81	233	582	1098	1744	2198	(9)
(10)		CDD50	1128	0	0	0	0	44	313	464	263	43	1	0	0	(10)
(11)		CDD65	145	0	0	0	0	2	43	80	20	0	0	0	0	(11)
(12)		CDH74	1160	0	0	0	0	14	421	587	137	1	0	0	0	(12)
(13)		CDH80	248	0	0	0	0	1	95	127	25	0	0	0	0	(13)
(14)	Wind	WSAvg	6.7	7.1	6.6	7.2	7.2	7.2	6.0	5.5	5.8	6.4	7.3	6.8	7.1	(14)
(15)	Precipitation	PrecAvg	22.2	1.3	1.1	1.0	1.1	1.8	2.6	2.3	2.6	2.5	2.4	2.0	1.6	(15)
(16)		PrecMax	30.8	2.8	1.9	2.0	2.3	4.3	5.6	4.2	4.5	4.3	3.9	3.6	3.3	(16)
(17)		PrecMin	16.3	0.0	0.4	0.3	0.4	0.6	0.9	0.3	0.5	0.9	1.4	0.7	0.7	(17)
(18)		PrecStd	3.2	0.6	0.4	0.4	0.5	0.9	1.1	1.1	1.0	0.9	0.6	0.7	0.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.5	31.1	41.9	56.1	75.6	86.9	87.8	83.5	70.1	54.0	36.0	30.6	(19)
(20)			MCWB	28.8	30.2	35.6	43.9	58.7	64.9	68.8	67.9	58.4	47.9	34.5	29.9	(20)
(21)		2%	DB	24.3	26.8	37.7	50.5	68.9	83.0	83.4	77.9	64.8	49.4	34.1	26.7	(21)
(22)			MCWB	23.7	25.4	32.7	41.1	54.2	63.7	67.1	65.0	56.0	45.9	33.2	25.9	(22)
(23)		5%	DB	19.1	22.6	34.3	46.1	63.8	79.2	80.1	73.6	60.9	45.8	32.5	23.1	(23)
(24)			MCWB	18.3	21.6	31.0	38.8	51.8	62.7	65.7	63.0	54.2	42.8	31.6	22.3	(24)
(25)		10%	DB	14.2	18.2	31.7	42.2	58.2	75.2	77.0	69.5	57.0	42.4	30.0	18.6	(25)
(26)			MCWB	13.4	17.1	29.2	36.4	48.6	61.5	64.7	61.5	52.1	39.8	29.0	17.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.9	30.5	36.4	45.8	59.7	68.9	72.4	69.6	60.0	49.7	34.9	29.8	(27)
(28)			MCD B	29.4	31.2	40.5	52.9	72.4	79.9	82.0	79.2	66.1	52.7	35.3	30.2	(28)
(29)		2%	WB	23.7	25.6	33.5	42.2	55.7	66.5	69.8	66.9	57.3	46.1	33.4	25.9	(29)
(30)			MCD B	24.3	26.6	36.5	49.5	66.4	76.4	78.7	75.0	63.2	48.9	33.9	26.5	(30)
(31)		5%	WB	18.5	21.5	31.5	39.3	52.7	64.7	68.2	64.4	55.1	43.3	31.7	22.4	(31)
(32)			MCD B	19.1	22.6	34.1	44.8	61.9	74.3	77.0	71.1	59.7	45.7	32.5	23.1	(32)
(33)		10%	WB	13.5	17.2	29.1	37.1	49.6	63.0	66.4	62.1	52.7	40.0	29.0	17.9	(33)
(34)			MCD B	14.2	18.1	31.7	41.4	57.9	72.8	74.0	68.3	56.7	42.1	30.1	18.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	13.0	15.5	18.4	16.5	16.0	17.1	16.3	14.2	11.6	8.7	11.5	13.2	(35)
(36)			MCD BR	13.1	16.0	16.6	20.5	24.0	25.0	21.9	20.7	18.2	11.0	8.4	15.0	(36)
(37)			MCWBR	13.2	15.8	13.4	13.4	12.6	9.9	9.2	9.8	10.3	8.3	8.6	14.9	(37)
(38)		5% WB	MCD BR	13.6	17.0	14.6	17.5	21.2	19.7	18.5	18.1	15.0	10.4	9.0	15.1	(38)
(39)			MCWBR	13.7	16.5	12.5	12.4	12.1	9.4	8.8	9.9	9.5	8.2	9.2	14.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.270	0.306	0.342	0.367	0.384	0.395	0.463	0.411	0.358	0.325	0.266	0.246	(40)	
(41)		taud	1.924	1.973	1.978	2.044	2.254	2.362	2.131	2.274	2.387	2.240	1.981	1.879	(41)	
(42)		Ebn at Noon	127	200	236	258	265	265	239	244	239	199	128	81	(42)	
(43)		Edh at Noon	15	27	38	43	38	35	43	35	26	21	14	10	(43)	
(44)	All-Sky Solar Radiation	RadAvg	122	404	845	1332	1560	1756	1619	1149	682	295	136	60	(44)	
(45)		RadStd	16	31	87	147	173	190	141	105	93	45	28	9	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page